2 Homework: Slope, Lines and Systems of Equations — Markscheme

1. [Maximum mark: 9]

(a) $m = \frac{4-1}{4-(-2)}$ **M1**

$m = \frac{3}{6} = \frac{1}{2}$ **A1**

(b) $y - 1 = \frac{1}{2}(x + 2)$ **M1**

$y = \frac{1}{2}x + 2$ **A1**

Accept $y - 4 = \frac{1}{2}(x - 4)$ *for the M1. FT from (a).*

(c) y-intercept: 2 **A1**

$0 = \frac{1}{2}x + 2$ **M1**

$x = -4$ **A1**

Accept $(0, 2)$ *and* $(-4, 0)$*. FT from (b).*

(d) $k = \frac{1}{2}(10) + 2$ **M1**

$k = 7$ **A1**

Accept the gradient method $\frac{k-4}{10-4} = \frac{1}{2}$ *for the M1. FT from (b).*

2. [Maximum mark: 9]

(a) $y = 0$: $2x = 12$, so $x = 6$ **A1**

$x = 0$: $3y = 12$, so $y = 4$ **A1**

Accept $(6, 0)$ *and* $(0, 4)$*.*

(b) $3y = -2x + 12$

$y = -\frac{2}{3}x + 4$ **M1**

$m = -\frac{2}{3}$ **A1**

Accept the gradient from the intercepts, $\frac{4-0}{0-6}$*, for the M1.*

(c) $m_2 = \frac{3}{2}$ **A1**

$y - 1 = \frac{3}{2}(x - 4)$ **M1A1**

Accept $y = \frac{3}{2}x - 5$ *or* $3x - 2y = 10$*. FT from (b): the gradient of* L_2 *is the negative reciprocal of the candidate's gradient.*

(d) $y = -\frac{2}{3}x - 2$ **A1A1**

A1 for the gradient $-\frac{2}{3}$*, A1 for the intercept* -2*. Accept* $2x + 3y = -6$*. FT from (b).*

3. [Maximum mark: 6]

(a) $3x + 2y = 16, \quad x - y = 2.$

$x = y + 2$

$3(y + 2) + 2y = 16$ **M1**

$5y + 6 = 16$

$y = 2$ **A1**

$x = 2 + 2 = 4$ **A1**

Accept elimination: $2x - 2y = 4$ *added to* $3x + 2y = 16$ *gives* $5x = 20$. *Correct values with no working earn 1 mark: the question asks for working.*

(b) $2x + 3y = 7, \quad 5x - 2y = -11.$

$4x + 6y = 14$

$15x - 6y = -33$ **M1**

$19x = -19$

$x = -1$ **A1**

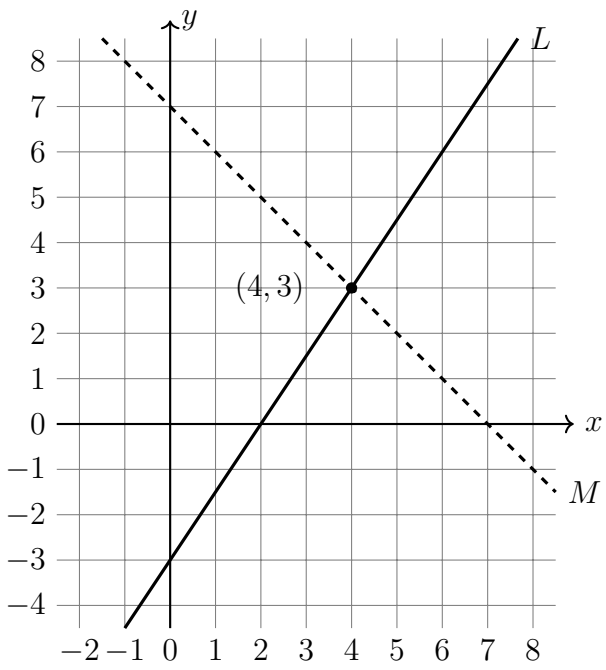
$2(-1) + 3y = 7$

$3y = 9$

$y = 3$ **A1**

M1 for multiplying to match the coefficients of one variable, or for a correct substitution. Correct values with no working earn 1 mark.

4. [Maximum mark: 9]



(a) $y = \frac{3}{2}x - 3$ **A1A1**

A1 for the gradient $\frac{3}{2}$, A1 for the intercept -3. The line passes through $(0, -3)$, $(2, 0)$ and $(4, 3)$.

(b) M through $(0, 7)$, $(7, 0)$ **A1A1**

A1 for each correct intercept, joined by a straight ruled line.

(c) $(4, 3)$, marked **A1A1**

A1 for the point marked, A1 for the ordered pair. FT from the candidate's line M.

(d) $x + \left(\frac{3}{2}x - 3\right) = 7$ **M1**

$\frac{5}{2}x - 3 = 7$

$\frac{5}{2}x = 10$

$x = 4$ **A1**

$y = 7 - 4 = 3$ **A1**

FT from the candidate's equation in (a). Accept $\frac{3}{2}x - 3 = 7 - x$ for the M1.